Classical theories of information science

信息度量
Information Measures

涂仕奎
上海交通大学
tushikui@sjtu.edu.cn
https://www.cs.sjtu.edu.cn/~tushikui/

Outline

1. Independence and Markov Chain
2. Shannon's Information Measures
   - Entropy, Conditional Entropy
   - Mutual Information, Conditional Mutual Information
3. Chain Rules
4. Information Divergence
5. The Basic Inequalities

1. Independence and Markov Chain

Independence and Markov Chain

Notations

X	discrete random variable taking values in $\mathcal{X}$
$\{p_X(x)\}$	probability distribution for X
$\mathcal{S}_X$	support of X

- If $\mathcal{S}_X = \mathcal{X}$, we say that p is strictly positive.
- Non-strictly positive distributions are dangerous.

Independence

Definition 2.1 Two random variables X and Y are independent, denoted by $X \perp Y$, if

$$p(x, y) = p(x)p(y)$$

for all x and y (i.e., for all $(x, y) \in \mathcal{X} \times \mathcal{Y}$).

Independence

Definition 2.1 Two random variables X and Y are independent, denoted by $X \perp Y$, if

$$p(x, y) = p(x)p(y)$$

for all x and y (i.e., for all $(x, y) \in \mathcal{X} \times \mathcal{Y}$).

Definition 2.2 (Mutual Independence) For $n \geq 3$, random variables $X_1, X_2, \cdots, X_n$ are mutually independent if

$$p(x_1, x_2, \cdots, x_n) = p(x_1)p(x_2) \cdots p(x_n)$$

for all $x_1, x_2, \cdots, x_n$.

Definition 2.3 (Pairwise Independence) For $n \geq 3$, random variables $X_1, X_2, \cdots, X_n$ are pairwise independent if X_i and X_j are independent for all $1 \leq i < j \leq n$.

Conditional Independence

Definition 2.4 (Conditional Independence) For random variables X, Y, and Z, X is independent of Z conditioning on Y, denoted by $X \perp Z|Y$, if

$$p(x, y, z)p(y) = p(x, y)p(y, z)$$

for all x, y, and z, or equivalently,

$$p(x, y, z) = \begin{cases} \frac{p(x,y)p(y,z)}{p(y)} = p(x, y)p(z|y) & \text{if } p(y) > 0 \\ 0 & \text{otherwise.} \end{cases}$$

Conditional Independence

Definition 2.4 (Conditional Independence) For random variables X, Y, and Z, X is independent of Z conditioning on Y, denoted by $X \perp Z|Y$, if

$$p(x, y, z)p(y) = p(x, y)p(y, z)$$

for all x, y, and z, or equivalently,

$$p(x, y, z) = \begin{cases} \frac{p(x,y)p(y,z)}{p(y)} = p(x, y)p(z|y) & \text{if } p(y) > 0 \\ 0 & \text{otherwise.} \end{cases}$$

Proposition 2.5 For random variables X, Y, and Z, $X \perp Z|Y$ if and only if

$$p(x, y, z) = a(x, y)b(y, z)$$

for all x, y, and z such that $p(y) > 0$.

Proposition 2.5 For random variables X, Y, and Z, $X \perp Z|Y$ if and only if

$$p(x,y,z) = a(x,y)b(y,z)$$

for all x, y, and z such that $p(y) > 0$.

$$p(x,y,z) = \begin{cases} \frac{p(x,y)p(y,z)}{p(y)} = p(x,y)p(z|y) & \text{if } p(y) > 0 \\ 0 & \text{otherwise.} \end{cases} \tag{2.4}$$

Proof. The 'only if' part follows immediately from the definition of conditional independence in (2.4), so we will only prove the 'if' part. Assume

$$p(x,y,z) = a(x,y)b(y,z) \tag{2.6}$$

for all x, y, and z such that $p(y) > 0$. Then for such x, y, and z, we have

$$p(x,y) = \sum_z p(x,y,z) = \sum_z a(x,y)b(y,z) = a(x,y)\sum_z b(y,z) \tag{2.7}$$

and

$$p(y,z) = \sum_x p(x,y,z) = \sum_x a(x,y)b(y,z) = b(y,z)\sum_x a(x,y). \tag{2.8}$$

Furthermore,

$$p(y) = \sum_z p(y,z) = \left(\sum_x a(x,y)\right)\left(\sum_z b(y,z)\right) > 0. \tag{2.9}$$

Proposition 2.5 For random variables X, Y, and Z, $X \perp Z|Y$ if and only if

$$p(x,y,z) = a(x,y)b(y,z)$$

for all x, y, and z such that $p(y) > 0$.

$$p(x,y,z) = \begin{cases} \frac{p(x,y)p(y,z)}{p(y)} = p(x,y)p(z|y) & \text{if } p(y) > 0 \\ 0 & \text{otherwise.} \end{cases} \tag{2.4}$$

Proof. The 'only if' part follows immediately from the definition of conditional independence in (2.4), so we will only prove the 'if' part. Assume

$$p(x,y,z) = a(x,y)b(y,z) \tag{2.6}$$

for all x, y, and z such that $p(y) > 0$. Then for such x, y, and z, we have

$$p(x,y) = \sum_z p(x,y,z) = \sum_z a(x,y)b(y,z) = a(x,y)\sum_z b(y,z) \tag{2.7}$$

and

$$p(y,z) = \sum_x p(x,y,z) = \sum_x a(x,y)b(y,z) = b(y,z)\sum_x a(x,y). \tag{2.8}$$

Furthermore,

$$p(y) = \sum_z p(y,z) = \left(\sum_x a(x,y)\right)\left(\sum_z b(y,z)\right) > 0. \tag{2.9}$$

Therefore,

$$\frac{p(x,y)p(y,z)}{p(y)} = \frac{\left(a(x,y)\sum_z b(y,z)\right)\left(b(y,z)\sum_x a(x,y)\right)}{\left(\sum_x a(x,y)\right)\left(\sum_z b(y,z)\right)} \tag{2.10}$$

$$= a(x,y)b(y,z) \tag{2.11}$$

$$= p(x,y,z). \tag{2.12}$$

For x, y, and z such that $p(y) = 0$, since

$$0 \le p(x,y,z) \le p(y) = 0, \tag{2.13}$$

we have

$$p(x,y,z) = 0. \tag{2.14}$$

Hence, $X \perp Z|Y$ according to (2.4). The proof is accomplished. □

Markov Chain

Definition 2.6 (Markov Chain). *For random variables $X_1, X_2, \cdots, X_n$, where $n \geq 3$, $X_1 \to X_2 \to \cdots \to X_n$ forms a Markov chain if*

$$\begin{aligned} & p(x_1, x_2, \cdots, x_n)p(x_2)p(x_3)\cdots p(x_{n-1}) \\ &= p(x_1, x_2)p(x_2, x_3)\cdots p(x_{n-1}, x_n) \end{aligned} \tag{2.15}$$

for all $x_1, x_2, \cdots, x_n$, or equivalently,

$$p(x_1, x_2, \cdots, x_n) =$$

$$\begin{cases} p(x_1, x_2)p(x_3|x_2)\cdots p(x_n|x_{n-1}) & \text{if } p(x_2), p(x_3), \cdots, p(x_{n-1}) > 0 \\ 0 & \text{otherwise.} \end{cases} \tag{2.16}$$

We note that $X \perp Z|Y$ is equivalent to the Markov chain $X \to Y \to Z$.

Markov Chain

$$p(x_1, x_2, \cdots, x_n)p(x_2)p(x_3)\cdots p(x_{n-1}) = p(x_1, x_2)p(x_2, x_3)\cdots p(x_{n-1}, x_n) \tag{2.15}$$

Proposition 2.7. *$X_1 \to X_2 \to \cdots \to X_n$ forms a Markov chain if and only if $X_n \to X_{n-1} \to \cdots \to X_1$ forms a Markov chain.*

Proof. This follows directly from the symmetry in the definition of a Markov chain in (2.15). □

Markov Chain

$$p(x_1, x_2, \cdots, x_n)p(x_2)p(x_3)\cdots p(x_{n-1}) = p(x_1, x_2)p(x_2, x_3)\cdots p(x_{n-1}, x_n) \tag{2.15}$$

Proposition 2.8. *$X_1 \to X_2 \to \cdots \to X_n$ forms a Markov chain if and only if*

$$\begin{aligned} &X_1 \to X_2 \to X_3 \\ &(X_1, X_2) \to X_3 \to X_4 \\ &\qquad\qquad \vdots \\ &(X_1, X_2, \cdots, X_{n-2}) \to X_{n-1} \to X_n \end{aligned} \tag{2.17}$$

form Markov chains.

Markov Chain

Proposition 2.5. *For random variables X, Y, and Z, $X \perp Z|Y$ if and only if*

$$p(x, y, z) = a(x, y)b(y, z) \tag{2.5}$$

for all x, y, and z such that $p(y) > 0$.

Proposition 2.9. *$X_1 \to X_2 \to \cdots \to X_n$ forms a Markov chain if and only if*

$$p(x_1, x_2, \cdots, x_n) = f_1(x_1, x_2) f_2(x_2, x_3) \cdots f_{n-1}(x_{n-1}, x_n) \tag{2.18}$$

for all $x_1, x_2, \cdots, x_n$ such that $p(x_2), p(x_3), \cdots, p(x_{n-1}) > 0$.

Note that Proposition 2.9 is a generalization of Proposition 2.5.

Markov subchains

Proposition 2.9. *$X_1 \to X_2 \to \cdots \to X_n$ forms a Markov chain if and only if*

$$p(x_1, x_2, \cdots, x_n) = f_1(x_1, x_2) f_2(x_2, x_3) \cdots f_{n-1}(x_{n-1}, x_n) \tag{2.18}$$

for all $x_1, x_2, \cdots, x_n$ such that $p(x_2), p(x_3), \cdots, p(x_{n-1}) > 0$.

Proposition 2.10 (Markov subchains). *Let $\mathcal{N}_n = \{1, 2, \cdots, n\}$ and let $X_1 \to X_2 \to \cdots \to X_n$ form a Markov chain. For any subset α of $\mathcal{N}_n$, denote $(X_i, i \in \alpha)$ by X_α. Then for any disjoint subsets $\alpha_1, \alpha_2, \cdots, \alpha_m$ of $\mathcal{N}_n$ such that*

$$k_1 < k_2 < \cdots < k_m \tag{2.19}$$

for all $k_j \in \alpha_j$, $j = 1, 2, \cdots, m$,

$$X_{\alpha_1} \to X_{\alpha_2} \to \cdots \to X_{\alpha_m} \tag{2.20}$$

forms a Markov chain. That is, a subchain of $X_1 \to X_2 \to \cdots \to X_n$ is also a Markov chain.

Markov subchains

Example 2.11 Let $X_1 \to X_2 \to \cdots \to X_{10}$ form a Markov chain and $\alpha_1 = \{1, 2\}$, $\alpha_2 = \{4\}$, $\alpha_3 = \{6, 8\}$, and $\alpha_4 = \{10\}$ be subsets of $\mathcal{N}_{10}$. Then Proposition 2.10 says that

$$(X_1, X_2) \to X_4 \to (X_6, X_8) \to X_{10}$$

also forms a Markov chain.

Proposition 2.12 Let X_1, X_2, X_3, and X_4 be random variables such that $p(x_1, x_2, x_3, x_4)$ is strictly positive. Then

$$\left.\begin{array}{l} X_1 \perp X_4 | (X_2, X_3) \\ X_1 \perp X_3 | (X_2, X_4) \end{array}\right\} \Rightarrow X_1 \perp (X_3, X_4) | X_2.$$

- Not true if p is not strictly positive
- Let $X_1 = Y$, $X_2 = Z$, and $X_3 = X_4 = (Y, Z)$, where $Y \perp Z$
- Then $X_1 \perp X_4 | (X_2, X_3)$, $X_1 \perp X_3 | (X_2, X_4)$, but $X_1 \not\perp (X_3, X_4) | X_2$.
- p is not strictly positive because $p(x_1, x_2, x_3, x_4) = 0$ if $x_3 \neq (x_1, x_2)$ or $x_4 \neq (x_1, x_2)$.

Proposition 2.12. *Let X_1, X_2, X_3, and X_4 be random variables such that $p(x_1, x_2, x_3, x_4)$ is strictly positive. Then*

$$\left.\begin{array}{l} X_1 \perp X_4 | (X_2, X_3) \\ X_1 \perp X_3 | (X_2, X_4) \end{array}\right\} \Rightarrow X_1 \perp (X_3, X_4) | X_2. \tag{2.22}$$

Proof

Proof. If $X_1 \perp X_4 | (X_2, X_3)$, then

$$p(x_1, x_2, x_3, x_4) = \frac{p(x_1, x_2, x_3) p(x_2, x_3, x_4)}{p(x_2, x_3)}. \tag{2.23}$$

On the other hand, if $X_1 \perp X_3 | (X_2, X_4)$, then

$$p(x_1, x_2, x_3, x_4) = \frac{p(x_1, x_2, x_4) p(x_2, x_3, x_4)}{p(x_2, x_4)}. \tag{2.24}$$

Equating (2.23) and (2.24), we have

$$p(x_1, x_2, x_3) = \frac{p(x_2, x_3) p(x_1, x_2, x_4)}{p(x_2, x_4)}. \tag{2.25}$$

Proposition 2.12. *Let X_1, X_2, X_3, and X_4 be random variables such that $p(x_1, x_2, x_3, x_4)$ is strictly positive. Then*

$$\left.\begin{array}{l} X_1 \perp X_4 | (X_2, X_3) \\ X_1 \perp X_3 | (X_2, X_4) \end{array}\right\} \Rightarrow X_1 \perp (X_3, X_4) | X_2. \tag{2.22}$$

Proof

Proof. If $X_1 \perp X_4 | (X_2, X_3)$, then

$$p(x_1, x_2, x_3, x_4) = \frac{p(x_1, x_2, x_3) p(x_2, x_3, x_4)}{p(x_2, x_3)}. \tag{2.23}$$

On the other hand, if $X_1 \perp X_3 | (X_2, X_4)$, then

$$p(x_1, x_2, x_3, x_4) = \frac{p(x_1, x_2, x_4) p(x_2, x_3, x_4)}{p(x_2, x_4)}. \tag{2.24}$$

Equating (2.23) and (2.24), we have

$$p(x_1, x_2, x_3) = \frac{p(x_2, x_3) p(x_1, x_2, x_4)}{p(x_2, x_4)}. \tag{2.25}$$

Therefore,

$$p(x_1, x_2) = \sum_{x_3} p(x_1, x_2, x_3) \tag{2.26}$$

$$= \sum_{x_3} \frac{p(x_2, x_3) p(x_1, x_2, x_4)}{p(x_2, x_4)} \tag{2.27}$$

$$= \frac{p(x_2) p(x_1, x_2, x_4)}{p(x_2, x_4)}, \tag{2.28}$$

or

$$\frac{p(x_1, x_2, x_4)}{p(x_2, x_4)} = \frac{p(x_1, x_2)}{p(x_2)}. \tag{2.29}$$

Hence from (2.24),

$$p(x_1, x_2, x_3, x_4) = \frac{p(x_1, x_2, x_4) p(x_2, x_3, x_4)}{p(x_2, x_4)} = \frac{p(x_1, x_2) p(x_2, x_3, x_4)}{p(x_2)} \tag{2.30}$$

for all x_1, x_2, x_3, and x_4, i.e., $X_1 \perp (X_3, X_4) | X_2$. □

2. Shannon's Information Measures

- Entropy
- Conditional entropy
- Mutual information
- Conditional mutual information

Entropy

Definition 2.13 The entropy $H(X)$ of a random variable X is defined as

$$H(X) = -\sum_x p(x) \log p(x).$$

- Convention: summation is taken over $\mathcal{S}_X$.
- When the base of the logarithm is α, write $H(X)$ as $H_\alpha(X)$.
- Entropy measures the uncertainty of a discrete random variable.
- The unit for entropy is

bit	if $\alpha = 2$
nat	if $\alpha = e$
D-it	if $\alpha = D$

- $H(X)$ depends only on the distribution of X but not on the actual value taken by X, hence also write $H(p)$.
- A bit in information theory is different from a bit in computer science.

Entropy as Expectation

- Convention

$$Eg(X) = \sum_x p(x)g(x)$$

  where summation is over $\mathcal{S}_X$.

- Linearity

$$E[f(X) + g(X)] = Ef(X) + Eg(X)$$

- Can write

$$H(X) = -E \log p(X) = -\sum_x p(x) \log p(x)$$

- In probability theory, when $Eg(X)$ is considered, usually $g(x)$ depends only on the value of x but not on $p(x)$.

Binary Entropy Function

- For $0 \leq \gamma \leq 1$, define the binary entropy function

$$h_b(\gamma) = -\gamma \log \gamma - (1-\gamma) \log(1-\gamma)$$

  with the convention $0 \log 0 = 0$.

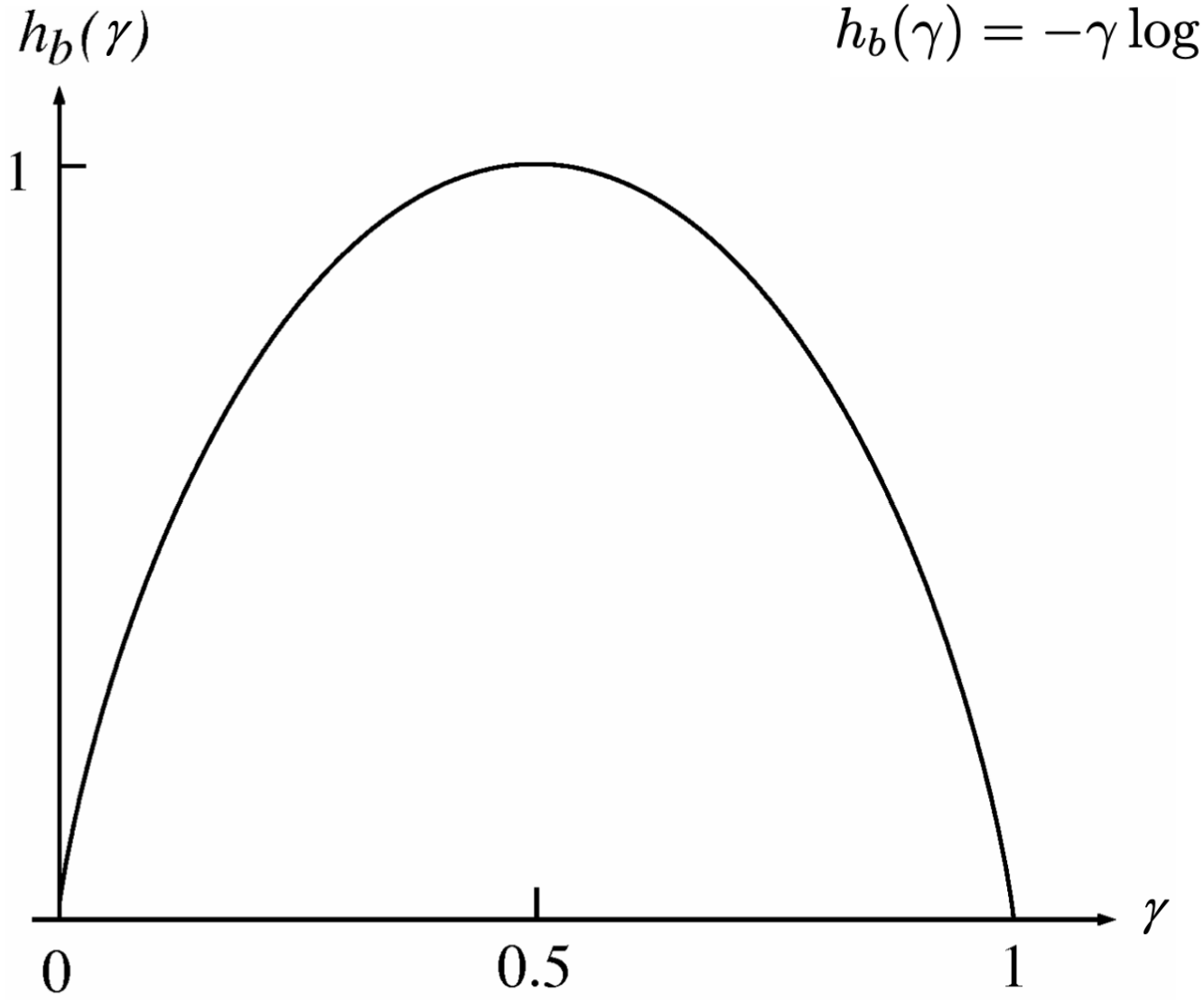
- For $X \sim \{\gamma, 1-\gamma\}$,

$$H(X) = h_b(\gamma).$$

- $h_b(\gamma)$ achieves the maximum value 1 when $\gamma = \frac{1}{2}$.

Binary Entropy Function

$$h_b(\gamma) = -\gamma \log \gamma - (1-\gamma) \log(1-\gamma)$$



Joint entropy, conditional entropy

Definition 2.14. *The joint entropy $H(X,Y)$ of a pair of random variables X and Y is defined as*

$$H(X,Y) = -\sum_{x,y} p(x,y) \log p(x,y) = -E \log p(X,Y). \qquad (2.40)$$

Joint entropy, conditional entropy

Definition 2.14 The joint entropy $H(X, Y)$ of a pair of random variables X and Y is defined as

$$H(X, Y) = -\sum_{x,y} p(x, y) \log p(x, y) = -E \log p(X, Y).$$

Definition 2.15 For random variables X and Y, the conditional entropy of Y given X is defined as

$$H(Y|X) = -\sum_{x,y} p(x, y) \log p(y|x) = -E \log p(Y|X).$$

Conditional entropy

- Write

$$H(Y|X) = \sum_x p(x) \left[-\sum_y p(y|x) \log p(y|x) \right].$$

- The inner sum is the entropy of Y conditioning on a fixed $x \in \mathcal{S}_X$, denoted by $H(Y|X = x)$.

- Thus

$$H(Y|X) = \sum_x p(x) H(Y|X = x),$$

- Similarly,

$$H(Y|X, Z) = \sum_z p(z) H(Y|X, Z = z),$$

where

$$H(Y|X, Z = z) = -\sum_{x,y} p(x, y|z) \log p(y|x, z).$$

Relationships

Proposition 2.16.

$$H(X,Y) = H(X) + H(Y|X) \tag{2.47}$$

and

$$H(X,Y) = H(Y) + H(X|Y). \tag{2.48}$$

Proof Consider

$$\begin{aligned} H(X,Y) &= -E\log p(X,Y) \\ &\overset{a)}{=} -E\log[p(X)p(Y|X)] \\ &\overset{b)}{=} -E\log p(X) - E\log p(Y|X) \\ &= H(X) + H(Y|X). \end{aligned}$$

a) summation is over $\mathcal{S}_{XY}$

b) linearity of expectation

Mutual Information

Definition 2.17 For random variables X and Y, the mutual information between X and Y is defined as

$$I(X;Y) = \sum_{x,y} p(x,y) \log \frac{p(x,y)}{p(x)p(y)} = E \log \frac{p(X,Y)}{p(X)p(Y)}.$$

Remark $I(X;Y)$ is symmetrical in X and Y.

Mutual Information

Proposition 2.18 The mutual information between a random variable X and itself is equal to the entropy of X, i.e., $I(X;X) = H(X)$.

Proof. This can be seen by considering

$$I(X;X) = E \log \frac{p(X)}{p(X)^2} \tag{2.54}$$

$$= -E \log p(X) \tag{2.55}$$

$$= H(X). \tag{2.56}$$

The proposition is proved. □

Remark The entropy of X is sometimes called the self-information of X.

Mutual Information

$$I(X;Y) = \sum_{x,y} p(x,y) \log \frac{p(x,y)}{p(x)p(y)} = E \log \frac{p(X,Y)}{p(X)p(Y)}. \quad (2.53)$$

Proposition 2.19

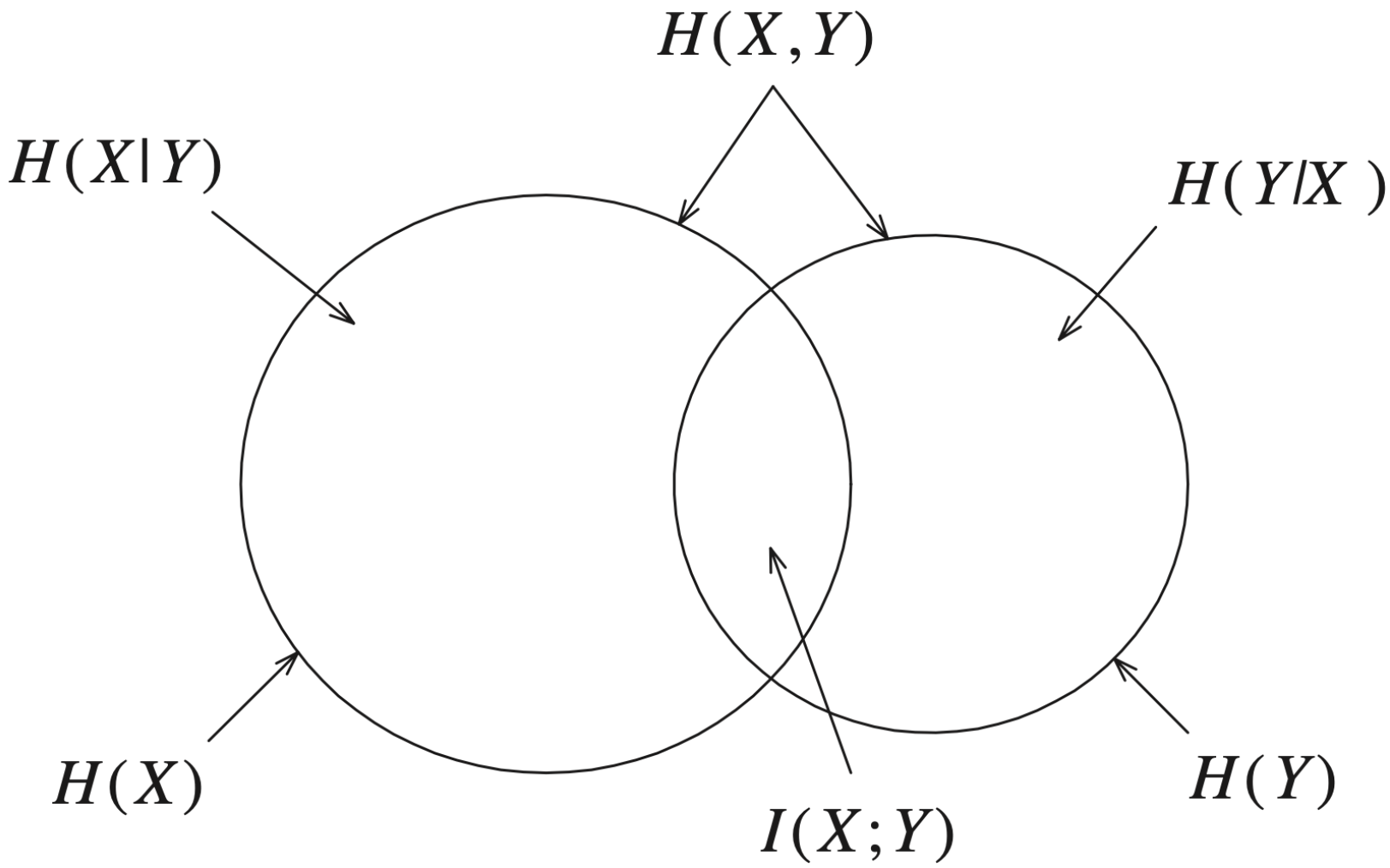
$$\begin{aligned} I(X;Y) &= H(X) - H(X|Y), \\ I(X;Y) &= H(Y) - H(Y|X), \end{aligned}$$

and

$$I(X;Y) = H(X) + H(Y) - H(X,Y),$$

provided that all the entropies and conditional entropies are finite.

Information Diagram



Conditional Mutual Information

Definition 2.20 For random variables X, Y and Z, the mutual information between X and Y conditioning on Z is defined as

$$I(X;Y|Z) = \sum_{x,y,z} p(x,y,z) \log \frac{p(x,y|z)}{p(x|z)p(y|z)} = E \log \frac{p(X,Y|Z)}{p(X|Z)p(Y|Z)}.$$

Remark $I(X;Y|Z)$ is symmetrical in X and Y.

Similar to entropy, we have

$$I(X;Y|Z) = \sum_{z} p(z) I(X;Y|Z=z),$$

where

$$I(X;Y|Z=z) = \sum_{x,y} p(x,y|z) \log \frac{p(x,y|z)}{p(x|z)p(y|z)}.$$

Propositions

Proposition 2.21 The mutual information between a random variable X and itself conditioning on a random variable Z is equal to the conditional entropy of X given Z, i.e., $I(X;X|Z) = H(X|Z)$.

Proposition 2.22

$$\begin{aligned} I(X;Y|Z) &= H(X|Z) - H(X|Y,Z), \\ I(X;Y|Z) &= H(Y|Z) - H(Y|X,Z), \end{aligned}$$

and

$$I(X;Y|Z) = H(X|Z) + H(Y|Z) - H(X,Y|Z),$$

provided that all the conditional entropies are finite.

Remark: All Shannon's information measures are special cases of conditional mutual information.

Continuity of Shannon's Information Measures for Fixed Finite Alphabets

- All Shannon's information measures are continuous when the alphabets are fixed and finite.

- For countable alphabets, Shannon's information measures are everywhere discontinuous.

Continuity

Definition 2.23 Let p and q be two probability distributions on a common alphabet $\mathcal{X}$. The variational distance between p and q is defined as

$$V(p,q) = \sum_{x \in \mathcal{X}} |p(x) - q(x)|.$$

The entropy function is continuous at p if

$$\lim_{p' \to p} H(p') = H\left(\lim_{p' \to p} p'\right) = H(p),$$

or equivalently, for any $\epsilon > 0$, there exists $\delta > 0$ such that

$$|H(p) - H(q)| < \epsilon$$

for all $q \in \mathcal{P}_{\mathcal{X}}$ satisfying

$$V(p,q) < \delta,$$

3. Chain Rules

Chain rules for entropy

Proposition 2.16.

$$H(X,Y) = H(X) + H(Y|X) \tag{2.47}$$

and

$$H(X,Y) = H(Y) + H(X|Y). \tag{2.48}$$

Proposition 2.24 (Chain Rule for Entropy).

$$H(X_1, X_2, \cdots, X_n) = \sum_{i=1}^{n} H(X_i|X_1, \cdots, X_{i-1}). \tag{2.85}$$

Proof. The chain rule for $n = 2$ has been proved in Proposition 2.16. We prove the chain rule by induction on n. Assume (2.85) is true for $n = m$, where $m \geq 2$. Then

$$\begin{aligned}
&H(X_1, \cdots, X_m, X_{m+1}) \\
&= H(X_1, \cdots, X_m) + H(X_{m+1}|X_1, \cdots, X_m) && (2.86)\\
&= \sum_{i=1}^{m} H(X_i|X_1, \cdots, X_{i-1}) + H(X_{m+1}|X_1, \cdots, X_m) && (2.87)\\
&= \sum_{i=1}^{m+1} H(X_i|X_1, \cdots, X_{i-1}), && (2.88)
\end{aligned}$$

where in (2.86) we have used (2.47) by letting $X = (X_1, \cdots, X_m)$ and $Y = X_{m+1}$, and in (2.87) we have used (2.85) for $n = m$. This proves the chain rule for entropy. □

Chain rules for entropy

Proposition 2.24 (Chain Rule for Entropy).

$$H(X_1, X_2, \cdots, X_n) = \sum_{i=1}^{n} H(X_i|X_1, \cdots, X_{i-1}). \tag{2.85}$$

Proposition 2.25 (Chain Rule for Conditional Entropy).

$$H(X_1, X_2, \cdots, X_n|Y) = \sum_{i=1}^{n} H(X_i|X_1, \cdots, X_{i-1}, Y). \tag{2.89}$$

Proof. The proposition can be proved by considering

$$\begin{aligned}
&H(X_1, X_2, \cdots, X_n|Y) \\
&= H(X_1, X_2, \cdots, X_n, Y) - H(Y) && (2.90)\\
&= H((X_1, Y), X_2, \cdots, X_n) - H(Y) && (2.91)\\
&= H(X_1, Y) + \sum_{i=2}^{n} H(X_i|X_1, \cdots, X_{i-1}, Y) - H(Y) && (2.92)\\
&= H(X_1|Y) + \sum_{i=2}^{n} H(X_i|X_1, \cdots, X_{i-1}, Y) && (2.93)\\
&= \sum_{i=1}^{n} H(X_i|X_1, \cdots, X_{i-1}, Y), && (2.94)
\end{aligned}$$

where (2.90) and (2.93) follow from Proposition 2.16, while (2.92) follows from Proposition 2.24.

Proof of chain rule for conditional entropy

Proposition 2.25 (Chain Rule for Conditional Entropy)

$$H(X_1, X_2, \cdots, X_n|Y) = \sum_{i=1}^{n} H(X_i|X_1, \cdots, X_{i-1}, Y).$$

Alternative Proof:

$$\begin{aligned}
& H(X_1, X_2, \cdots, X_n|Y) \\
= & \sum_y p(y) H(X_1, X_2, \cdots, X_n|Y = y) \\
= & \sum_y p(y) \sum_{i=1}^{n} H(X_i|X_1, \cdots, X_{i-1}, Y = y) \\
= & \sum_{i=1}^{n} \sum_y p(y) H(X_i|X_1, \cdots, X_{i-1}, Y = y) \\
= & \sum_{i=1}^{n} H(X_i|X_1, \cdots, X_{i-1}, Y),
\end{aligned}$$

Chain rule for mutual information

Proposition 2.26 (Chain Rule for Mutual Information).

$$I(X_1, X_2, \cdots, X_n; Y) = \sum_{i=1}^{n} I(X_i; Y | X_1, \cdots, X_{i-1}). \tag{2.99}$$

Proof. Consider

$$\begin{aligned} & I(X_1, X_2, \cdots, X_n; Y) \\ &= H(X_1, X_2, \cdots, X_n) - H(X_1, X_2, \cdots, X_n | Y) && (2.100) \\ &= \sum_{i=1}^{n} [H(X_i | X_1, \cdots, X_{i-1}) - H(X_i | X_1, \cdots, X_{i-1}, Y)] && (2.101) \\ &= \sum_{i=1}^{n} I(X_i; Y | X_1, \cdots, X_{i-1}), && (2.102) \end{aligned}$$

where in (2.101), we have invoked both Propositions 2.24 and 2.25. The chain rule for mutual information is proved. □

Chain rule for mutual information

Proposition 2.27 (Chain Rule for Conditional Mutual Information).
For random variables $X_1, X_2, \cdots, X_n$, Y, *and* Z,

$$I(X_1, X_2, \cdots, X_n; Y|Z) = \sum_{i=1}^{n} I(X_i; Y|X_1, \cdots, X_{i-1}, Z). \tag{2.103}$$

Proof. This is the conditional version of the chain rule for mutual information. The proof is similar to that for Proposition 2.25. The details are omitted. □

Thank you!